MAE 1001:
Blinking LEDs using the Raspberry Pi 4B+

Introduction to Mechanical and Aerospace Engineering

Course website: https://gwu-mae1001.github.io/fall2026/

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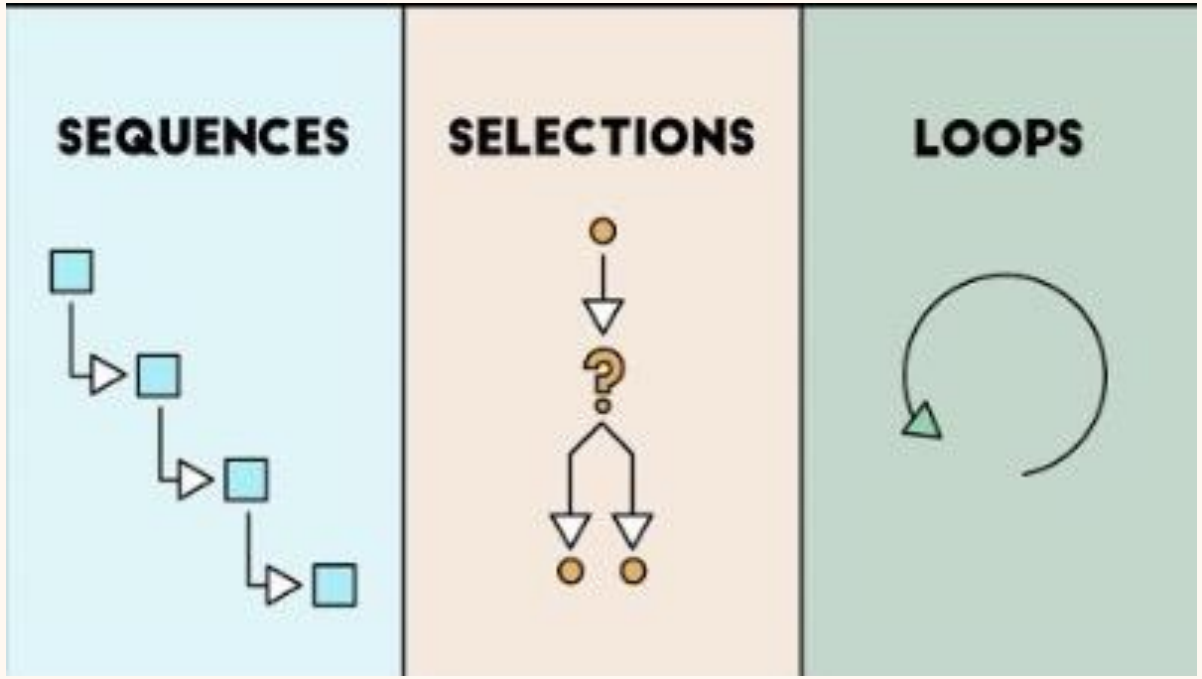
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Photo: Kartik Bulusu

Let’s get to know some programming paradigms

Sources:
Computer Science Basics: Sequences, Selections, and Loops: https://youtu.be/eSYeHlwDCNA



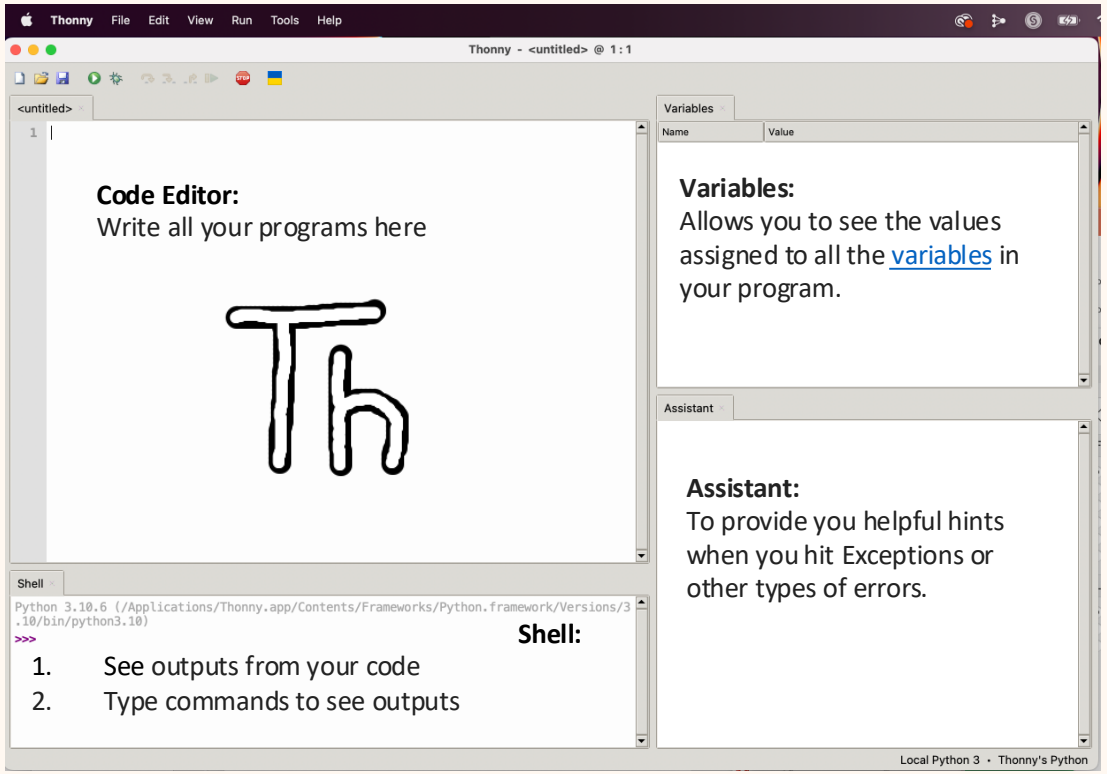
Sequences

Selections

Loops

Thonny integrated development environment (IDE)

Sources:
Thonny (IDE): https://en.wikipedia.org/wiki/Thonny
Thonny: The Beginner-Friendly Python Editor: https://realpython.com/python-thonny/



Allows you to run the code, i.e., "Do what I told you do do!".

Allows you to debug your code.
A bug is another name for a problem.

Allows you to open a file that already exists on your computer

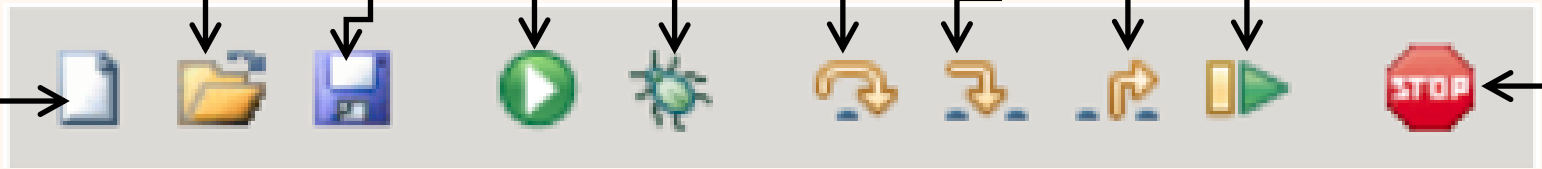
Save your code. Press this early and often.

Create a new file

The arrow icons allow you to run your programs step by step.
These icons are used after you press the bug icon

The resume icon allows you to return to play mode from debug mode.

The stop icon allows you to stop running your code



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Built-in function range()

Python's range()function
returns a sequence of numbers
works only with integers

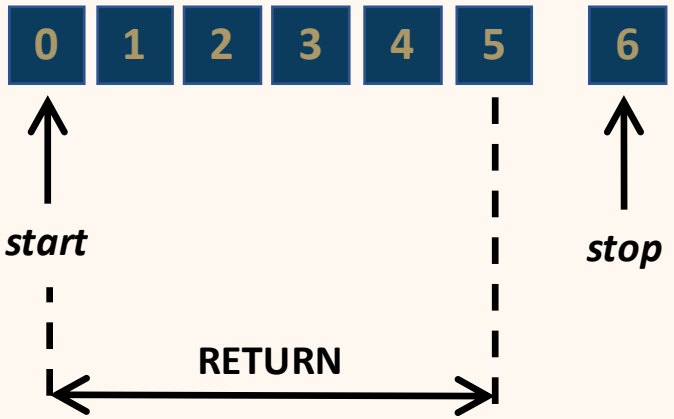
start:	at the value (default = 0)
step:	up or down at the increment value (default = 1)
stop:	at the value but not including it

range(stop)

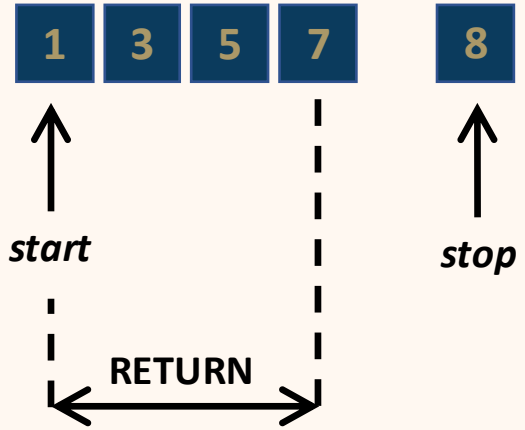
range(start, stop)

range(*start*, *stop*, *step*)

>>> range(6)



>>> range(1,8,2)



range()
- can be utilized in (for) loops
- to specify a range to iterate or do repetitions

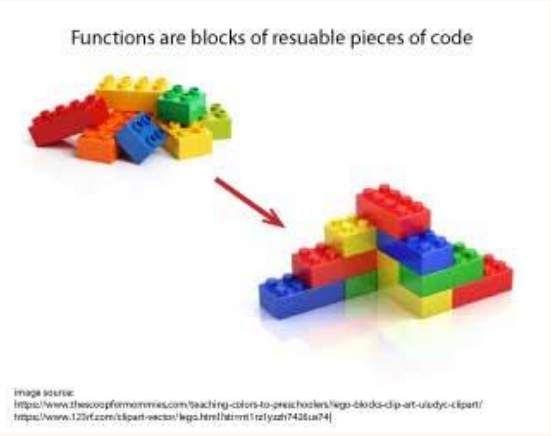
Demos

Syntax and Skeleton of a user-defined function

```
def name(parameters):

    statement
    statement
    . . .

    return value
```



Function name: Identifier by which it is called in the program

(Optional) **Arguments: values passed to the function**

Function Declaration: Starts with "def" that is not indented

```
def func_name(parameters):
```

Colon; Don't miss it!

Indentation: Tab or 4 spaces for each statement

```
    statement
    statement
    . . .

    return value
```

Body: Statements executed each time a function is called

(Optional) **return value: Can end function call and send data back to the main program**

Function definition

```
func_name()
```

Function call

Sources:
Codecademy: https://youtu.be/wxds6MAtUQ0

What is a ”loop” in computer programming ?

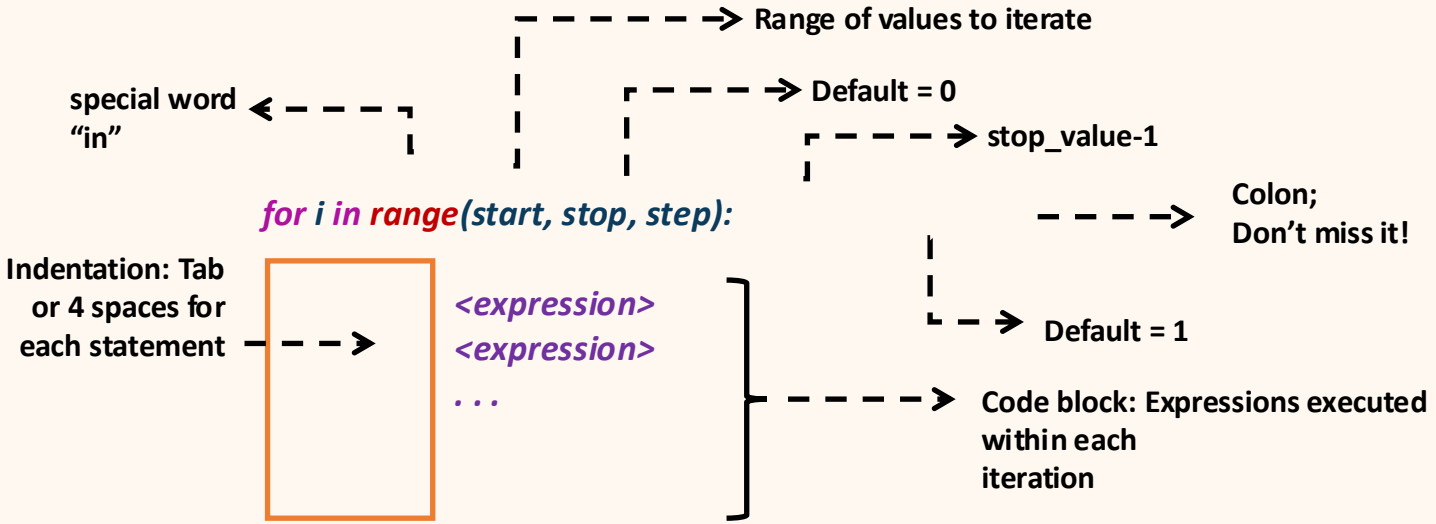


- Repeating routines
- From function to repeating functions
- Stopping condition

Skeleton of the for-loop

```
for i in range(start, stop, step):

        <expression>
        <expression>
        . . .
```



Animated example of the for-loop

Sources:
Animating Python #3- For Loop: https://youtu.be/EfJVnAHiR4s



Skeleton of the updated Python program written for Raspberry Pi

```
import library1 as name1
import RPi.GPIO as GPIO
import time
```

Import libraries that are relevant for interaction with the Raspberry Pi hardware such as GPIO pins, camera ports etc.

```
GPIO.setmode(GPIO.BOARD)

GPIO.setup(12, GPIO.OUT)
```

Set up GPIO pins as data outputs or inputs for sensors and actuators

```
def function_name(arguments):
        statement
        statement
        . . .
        return value
```

Create user-defined functions to modularize your code and make it easy to work.

Create user-defined functions to release the GPIO pins

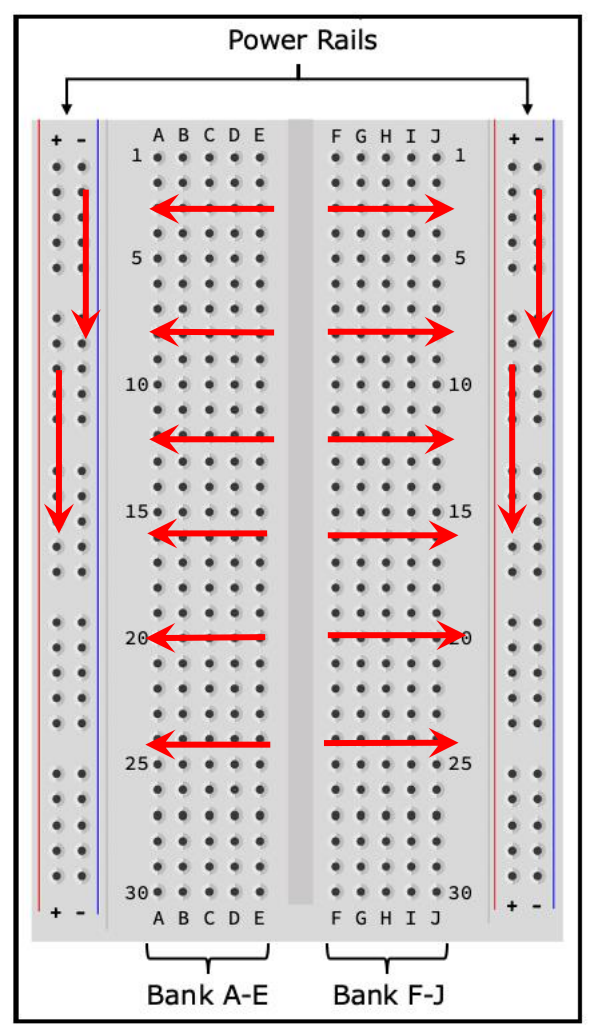
```
if __name__ == "__main__":
        try:
                function_name1()
        except KeyboardInterrupt:
                function_name2()
```

Create entry point into the program and pull all functions in

Create a keyboard-interrupt exception clause

Connect the Raspberry Pi Model 3 B+ (RPi) to a bread board

Sources of the breadboard image:
Gary Smart, Practical Python Programming for IoT









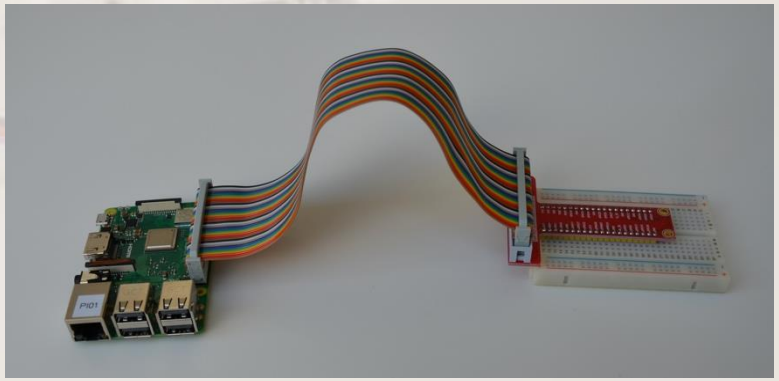
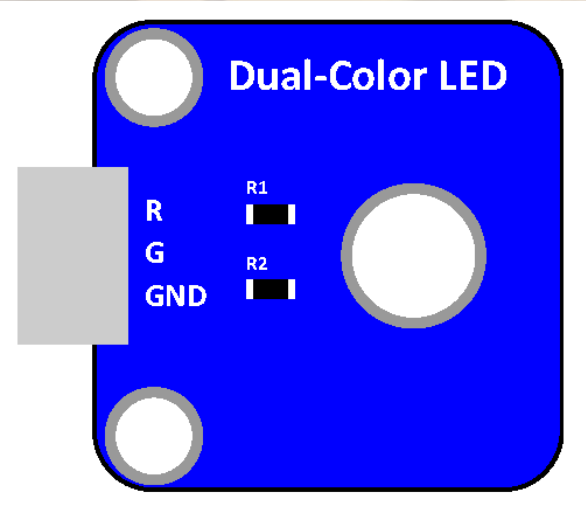


Photo: Kartik Bulusu

Know your Light Emitting Diode (LED)





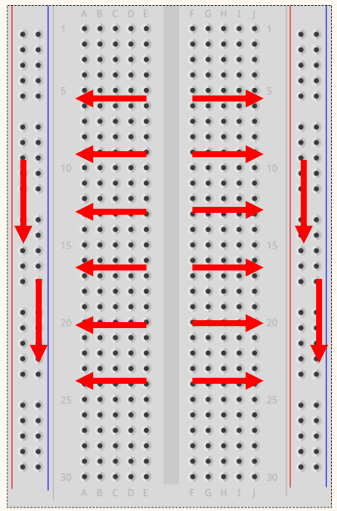
Source:
https://www.sunfounder.com/learn/lesson-1-dual-color-led-sensor-kit-v2-0-for-b.html

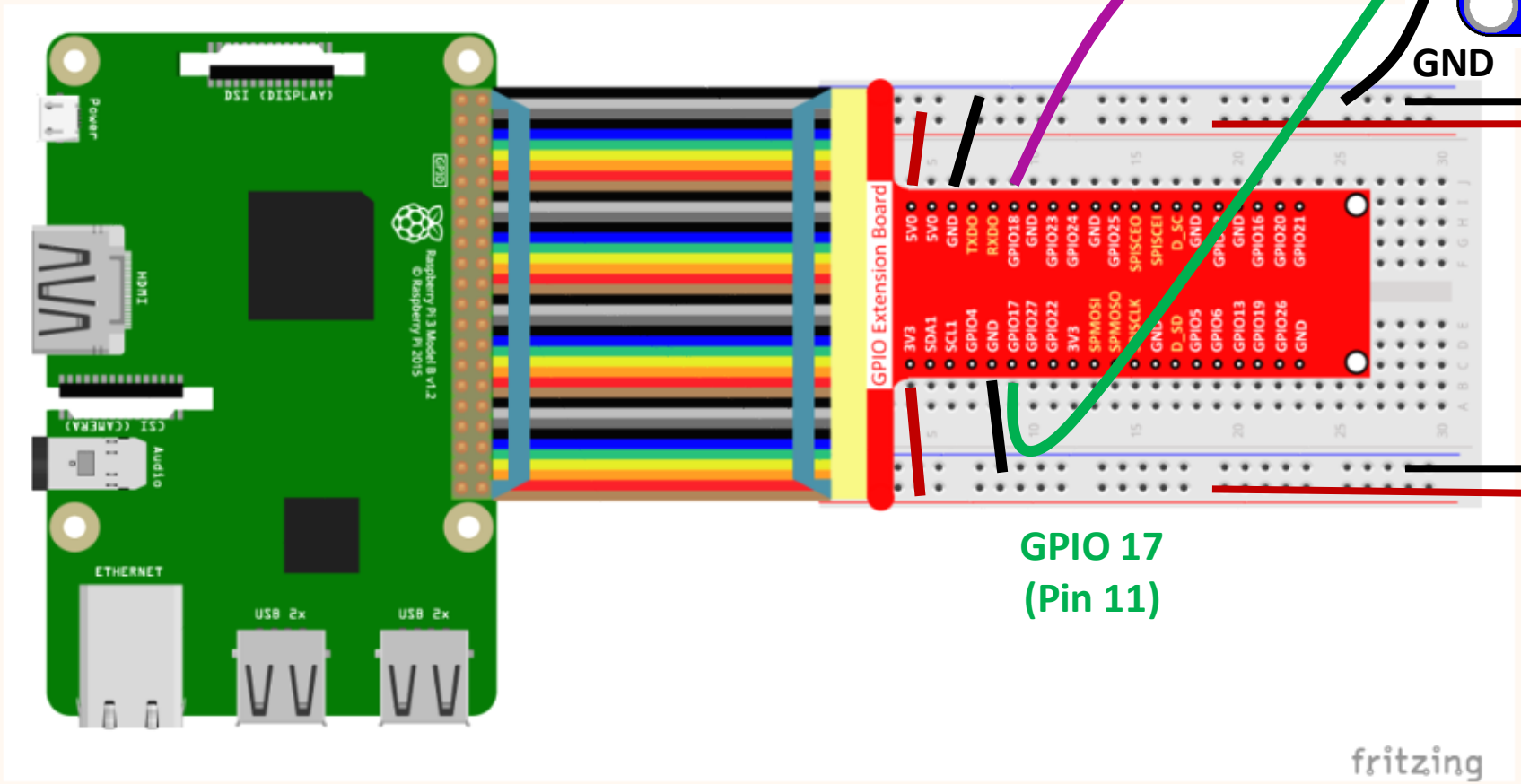
A dual-color light emitting diode (LED) is capable of emitting two different colors of light, typically red and green.

Application:
Variety of devices, such as televisions, digital cameras, and remote controls deploy these type LEDs.

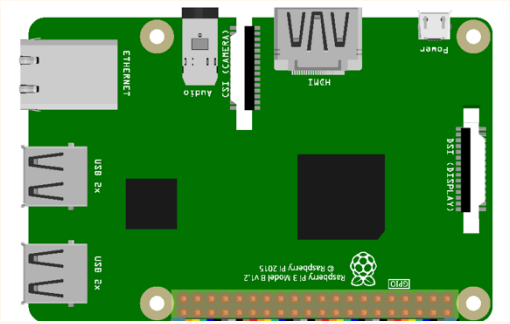
Connector:
3-pin anti-reverse cable

Light up an LED with your Raspberry Pi 3 B +



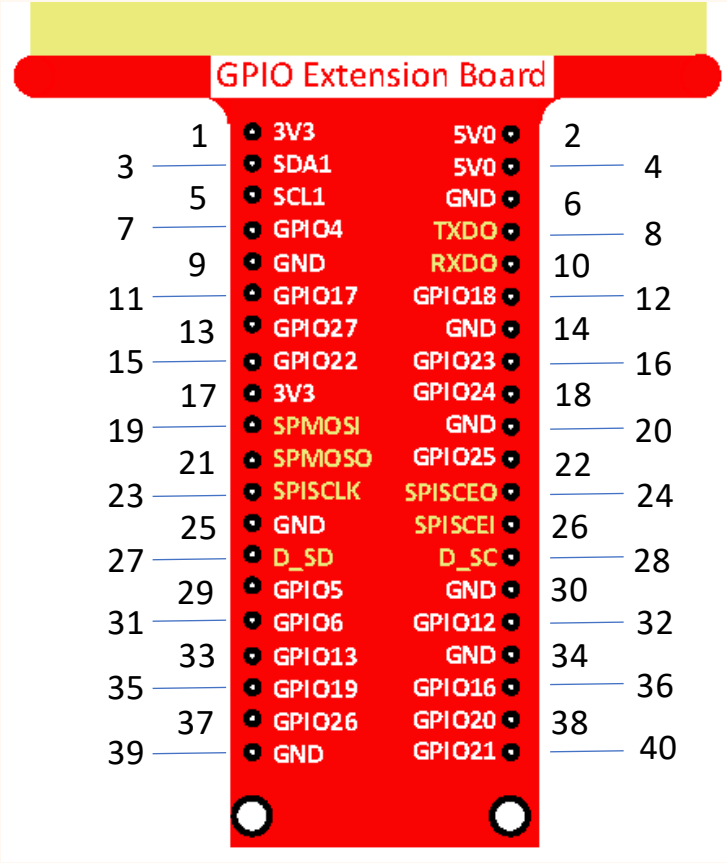


How a python code can light up your LED with Raspberry Pi Model 3 B+ (RPi)

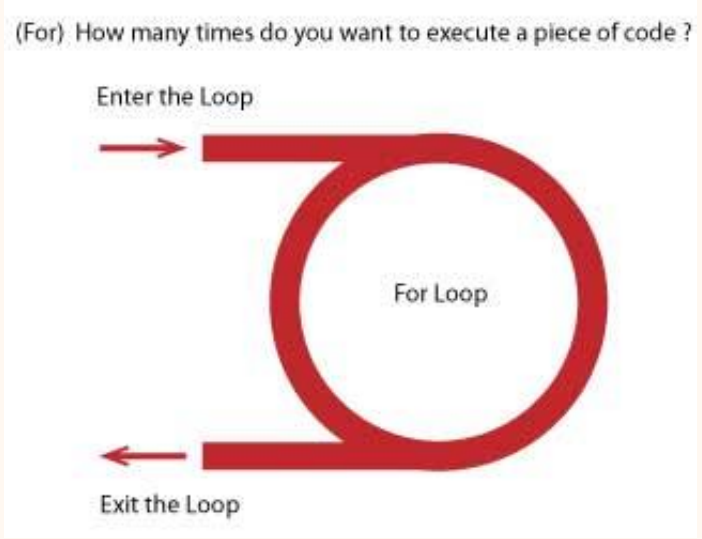


```
import RPi.GPIO as GPIO
import time
```

```
GPIO.setmode(GPIO.BOARD)
```




```
GPIO.setup(12, GPIO.OUT)
```

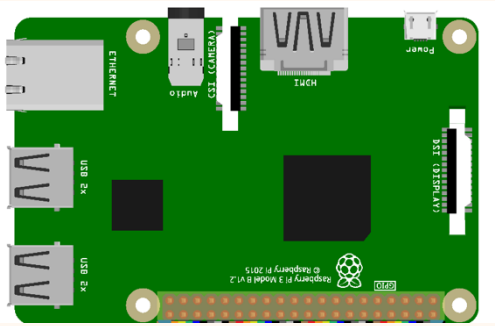



```
for i in range(0,15):

    GPIO.output(12, GPIO.HIGH)
    time.sleep(0.5)
    GPIO.output(12, GPIO.LOW)
    time.sleep(0.5)
    print(i)

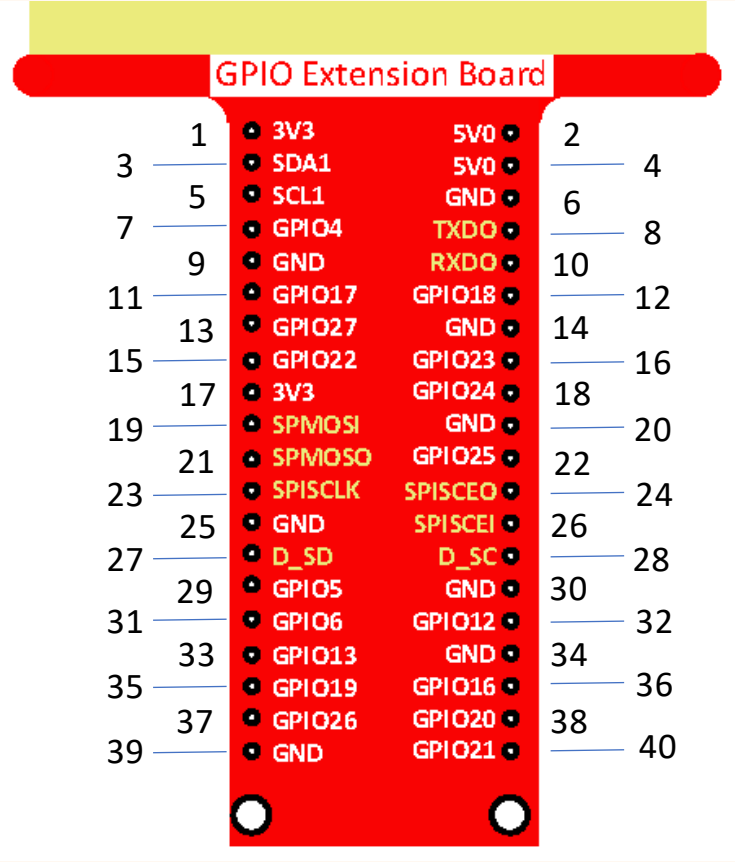
GPIO.cleanup()
```

Another python code to kick start your Raspberry Pi Model 3 B+ (RPi)



```
import RPi.GPIO as GPIO
import time
```

```
GPIO.setmode(GPIO.BOARD)
```




```
GPIO.setup(12, GPIO.OUT)
```

```
def loop():
        while True:
                GPIO.output(12, GPIO.HIGH)
        time.sleep(0.5)
            GPIO.output(12, GPIO.LOW)
        time.sleep(0.5)
```

```
def destroy():
    GPIO.output(12, GPIO.LOW)
            # Turn off all leds
    GPIO.cleanup()
```

```
if __name__ == "__main__":
    try:
        loop()
    except KeyboardInterrupt:
        destroy()
```



Photo: Kartik Bulusu

Someone should summarize what we learned today